Lowest Common Multiple (LCM)

1. Find $\text{lcm}(4, 6)$
2. Find $\text{lcm}(8, 12)$
3. Find $\text{lcm}(9, 15)$
4. Find $\text{lcm}(14, 21)$
5. Find $\text{lcm}(16, 20)$
6. Find $\text{lcm}(18, 24)$
7. Find $\text{lcm}(6, 8, 12)$
8. Find $\text{lcm}(10, 15, 25)$
9. Two lighthouse beacons flash simultaneously at midnight. Beacon A flashes every 12 seconds, and Beacon B flashes every 18 seconds. After how many seconds will they next flash together?
10. A bakery sells buns in packs of 8 and burgers in packs of 12. What is the smallest number of buns and burgers a customer must buy so that there is an equal number of each with none left over?

Solutions

1. **12**
   Explanation: List the positive multiples of each number until the first shared value appears:

$$\text{Multiples of } 4 : 4, 8, \mathbf{12}, 16, \ldots$$
$$\text{Multiples of } 6 : 6, \mathbf{12}, 18, 24, \ldots$$

   The smallest common multiple is 12.

2. **24**
   Explanation: List the positive multiples of 8 and 12:

$$\text{Multiples of } 8 : 8, 16, \mathbf{24}, 32, \ldots$$
$$\text{Multiples of } 12 : 12, \mathbf{24}, 36, \ldots$$

   Thus, $\text{lcm}(8, 12) = 24$.

3. **45**
   Explanation: Using prime factorisation:

$$9 = 3^2$$
$$15 = 3 \times 5$$

   Take the highest power of each prime factor present: $\text{lcm}(9, 15) = 3^2 \times 5 = 45$.

4. **42**
   Explanation: Using prime factorisation:

$$14 = 2 \times 7$$
$$21 = 3 \times 7$$

   Multiply the highest powers of all prime factors: $\text{lcm}(14, 21) = 2 \times 3 \times 7 = 42$.

5. **80**
   Explanation: Using prime factorisation:

$$16 = 2^4$$
$$20 = 2^2 \times 5$$

   Take the highest power of each prime factor: $\text{lcm}(16, 20) = 2^4 \times 5 = 16 \times 5 = 80$.

6. **72**
   Explanation: Using prime factorisation:

$$18 = 2 \times 3^2$$
$$24 = 2^3 \times 3$$

   Multiply the highest powers of each prime factor: $\text{lcm}(18, 24) = 2^3 \times 3^2 = 8 \times 9 = 72$.

7. **24**
   Explanation: Find the prime factorisation of each of the three numbers:

$$6 = 2 \times 3$$
$$8 = 2^3$$
$$12 = 2^2 \times 3$$

   Taking the highest power of each prime gives $\text{lcm}(6, 8, 12) = 2^3 \times 3 = 24$.

8. **150**
   Explanation: Find the prime factorisation of each number:

$$10 = 2 \times 5$$
$$15 = 3 \times 5$$
$$25 = 5^2$$

   Taking the highest power of each prime factor gives $\text{lcm}(10, 15, 25) = 2 \times 3 \times 5^2 = 6 \times 25 = 150$.

9. **36 seconds**
   Explanation: The beacons flash together again at the lowest common multiple of 12 and 18:

$$12 = 2^2 \times 3$$
$$18 = 2 \times 3^2$$

   Thus, $\text{lcm}(12, 18) = 2^2 \times 3^2 = 4 \times 9 = 36$. They will next flash together after 36 seconds.

10. **24 buns and 24 burgers (3 packs of buns and 2 packs of burgers)**
    Explanation: Find the lowest common multiple of 8 and 12:

$$8 = 2^3$$
$$12 = 2^2 \times 3$$

    Thus, $\text{lcm}(8, 12) = 2^3 \times 3 = 24$. To get 24 of each, the customer needs $\dfrac{24}{8} = 3$ packs of buns and $\dfrac{24}{12} = 2$ packs of burgers.